CSE-476: Introduction to Natural Language Processing

Quiz 1

1 Multiple Choice Questions (Single correct)

Instructions: Instructions: Select the correct option with a short explanation.

1. In the BIO tagging scheme for NER, what does the "I-" prefix indicate? **A.** Beginning of an entity **B.** Inside an entity **C.** Outside any entity **D.** Between two tokens

2. Which model considers only the observed words and ignores label transitions? **A.** It uses only words (observations) and ignores labels **B.** It is identical to an HMM **C.** It models transitions between labels but does not consider the words **D.** It models emissions P(word | label)

3. In a Hidden Markov Model (HMM) for NER, a transition probability is: **A.** P(label | previous label) **B.** P(label | word) **C.** P(word | previous word) **D.** P(word | label)

4. Which task assigns grammatical roles like subject and object from syntax? **A.** A task that identifies predicates in a sentence and labels roles of elements such as agent, theme, or location **B.** A process that assigns grammatical roles like subject and object based on syntactic structure **C.** A method that classifies named entities and links them to a knowledge base **D.** A technique that labels words with their parts of speech and morphological features

5. Which method compresses text by replacing consecutive repeats with a count and symbol? **A.** A technique that compresses text by replacing consecutive repeated characters with a count and a symbol **B.** A tokenization method that iteratively

merges the most frequent pair of symbols in a corpus **C.** A tokenization method that splits text based on whitespace and punctuation **D.** A character-level encoding scheme that assigns fixed-length binary codes to characters

6. Which task labels each token with a grammatical category like noun or verb?
   A. To label each token with a grammatical category **B.** To convert text into a sequence of symbols suitable for modeling **C.** To reduce vocabulary size to zero **D.** To assign probabilities to sentences

7. Which drawback claims this method produces too many tokens when segmenting text?
   A. It produces too many tokens **B.** It does not work well for languages without explicit word boundaries **C.** It cannot tokenize English at all **D.** It ignores punctuation entirely

8. Which task is commonly used to assign part-of-speech tags to words?
   A. Part-of-speech tagging **B.** Named Entity Recognition **C.** Dependency parsing **D.** Sentiment analysis

9. Which limitation claims that BIO tagging requires too many distinct labels?
   A. It cannot represent overlapping or nested entities **B.** It requires too many labels **C.** It does not support multiple entity types **D.** It ignores entity boundaries

10. Which trade-off is often discussed when balancing model speed
    A. Speed vs accuracy **B.** Tokenized sequence length vs vocabulary size **C.** Precision vs recall **D.** Memory vs computation